

WEEK6

Checkpairwithdifferencek

Problem Statement:

Given an array A of sorted integers and another non-negative integer k , find if there exists 2 indices i and j such that $A[i] - A[j] = k$, $i \neq j$.

Input Format

1. First line is number of test cases T . Following T lines contain:
2. N , followed by N integers of the array
3. The non-negative integer k

Output format

Print 1 if such a pair exists and 0 if it doesn't. Sample

Input:

1

3 1 3 5

4

Sample Output:

1

```

1  #include<stdio.h>
2  int main()
3  {
4      int t;
5      scanf("%d",&t);
6      while(t--)
7      {
8          int n;
9          scanf("%d",&n);
10         int a[n];
11         for(int i=0;i<n;i++)
12         {
13             scanf("%d",&a[i]);
14         }
15         int k;
16         scanf("%d",&k);
17         int flag = 0;
18         for (int i=0;i<n;i++)
19         {
20             for(int j =i+1;j<n;j++)
21             {
22                 if (a[i]-a[j]==k || a[j]-a[i]==k)
23                 {
24                     flag = 1;
25                     break;
26                 }
27             }
28             if(flag)break;
29         }
30         printf("%d\n",flag);
31     }
32 }
```

	Input	Expected	Got	
✓	1 3 1 3 5 4	1	1	✓
✓	1 3 1 3 5 99	0	0	✓

Passed all tests! ✓

Chocolates

Problem Statement:

Sam loves chocolates and starts buying them on the 1st day of the year. Each day of the year, x , is numbered from 1 to Y . On days when x is odd, Sam will buy x chocolates; on days when x is even, Sam will not purchase any chocolates.

Complete the code in the editor so that for each day N_i (where $1 \leq x \leq N \leq Y$) in array `arr`, the number of chocolates Sam purchased (during days 1 through N) is printed on a new line. This is a function-only challenge, so input is handled for you by the locked stub code in the editor.

Input Format

The program takes an array of integers as a parameter.

The locked code in the editor handles reading the following input from `stdin`, assembling it into an array of integers (`arr`), and calling `calculate(arr)`.

The first line of input contains an integer, T (the number of test cases). Each line i of the T subsequent lines describes the i th test case as an integer, N_i (the number of days).

Constraints

$$1 \leq T \leq 2 \times 10^5$$

$$1 \leq N \leq 2 \times 10^6$$

$$1 \leq x \leq N \leq Y$$

Output Format

For each test case, T_i in `arr`, your `calculate` method should print the total number of

chocolatesSampurchasedbydayNionanewline.

Sample Input 0

3

1

2

3

SampleOutput 0

1

1

4

```

1 #include<stdio.h>
2 int main()
3 {
4     int t;
5     scanf("%d", &t);
6     while(t--)
7     {
8         int n,c=0;
9         scanf("%d",&n);
10        for(int i = 0;i<=n;i++)
11        {
12            if(i%2 != 0)
13            {
14                c = c+ i;
15            }
16        }
17        printf("%d\n",c);
18    }
19 }

```

	Input	Expected	Got	
✓	3	1	1	✓
	1	1	1	
	2	4	4	
	3			
✓	10	1296	1296	✓
	71	2500	2500	
	100	1849	1849	
	86	729	729	
	54	400	400	
	40	25	25	
	9	1521	1521	
	77	25	25	
	9	49	49	
	13	2401	2401	
	98			

Passed all tests! ✓

Football Scores

Problem Statement:

The number of goals achieved by two football teams in matches in a league is given in the form of two lists. Consider:

- Football team A, has played three matches, and has scored {1,2,3} goals in each match respectively.
- Football team B, has played two matches, and has scored {2,4} goals in each match respectively.
- Your task is to compute, for each match of team B, the total number of matches of team A, where team A has scored less than or equal to the number of goals scored by team B in that match.

In the above case:

- For 2 goals scored by team B in its first match, team A has 2 matches with scores 1 and 2.
- For 4 goals scored by team B in its second match, team A has 3 matches with scores 1, 2 and 3.

Hence, the answer: {2, 3}.

Complete the code in the editor below. The program must return an array of positive integers, one for each $\text{maxes}[i]$ representing the total number of elements $\text{nums}[j]$ satisfying $\text{nums}[j] \leq \text{maxes}[i]$ where $0 \leq j < n$ and $0 \leq i < m$, in the given order. It has the following:

$\text{nums}[\text{nums}[0], \dots, \text{nums}[n-1]]$: first array of positive integers

$\text{maxes}[\text{maxes}[0], \dots, \text{maxes}[m-1]]$: second array of positive integers

Constraints:

$2 \leq n, m \leq 105, 1 \leq \text{nums}[j] \leq 109$, where $0 \leq j < n$, $1 \leq \text{maxes}[i] \leq 109$, where $0 \leq i < m$. Input Format For

Custom Testing

Input from stdin will be processed as follows and passed to the function. The first line contains an integer n , the number of elements in nums .

The next n lines each contain an integer describing $\text{nums}[j]$ where $0 \leq j < n$. The next line contains an integer m , the number of elements in maxes .

The next m lines each contain an integer describing $\text{maxes}[i]$ where $0 \leq i < m$. Sample

Input

4

1

4

2

4

2

3

5

SampleOutput

2

4

```
1 #include<stdio.h>
2 int main()
3 {
4     int a,b,ans;
5     scanf("%d",&a);
6     int c[a];
7     for (int i =0;i<a;i++)
8         scanf("%d",&c[i]);
9     scanf("%d",&b);
10    int d[b];
11    for(int i=0;i<b;i++)
12        scanf("%d",&d[i]);
13    for(int j=0;j<b;j++)
14    {
15        ans = 0;
16        for(int i =0;i<a;i++)
17        {
18            if(d[j]>=c[i])
19                ans++;
20        }
21        printf("%d\n",ans);
22    }
23 }
24 }
```

	Input	Expected	Got	
✓	4	2	2	✓
	1	4	4	
	4			
	2			
	4			
	2			
	3			
	5			
✓	5	1	1	✓
	2	0	0	
	10	3	3	
	5	4	4	
	4			
	8			
	4			
	3			
	1			
	7			
	8			

Passed all tests! ✓